LAB3: An electronic stethoscope and a defibrillator's ready light, and the radio's amplifier

Week 7 · 1 h 50 min, teams of three

Lecturer: Dr. Asem Alaa Teaching assistants: Sara Habo and Mikhail Nady Daoud

Last updated 29 October 2026

Goal

Part A builds two medical devices from lectures L05 and L06. An electronic stethoscope plays the heart's sounds into an earphone: a microphone in a small funnel pressed to the chest, L05's bench amplifier, then a follower that lets it drive an earphone of 32 Ω. A defibrillator's ready light waits until its capacitor holds enough charge: a Zener diode and a metal-oxide-semiconductor field-effect transistor (MOSFET) switch a light-emitting diode (LED) on once the capacitor passes their two voltages. Part B is milestone 2 (M2) of the medium-wave radio your team builds this term: board 1′, two common-emitter (CE) stages of lecture L05 and a detector, built on a breadboard for now; soldering waits for M3. It multiplies the station that M1's tank picks by several hundred before the detector, so that a station far too weak for M1's diode becomes a clear signal. Predict each result by hand, and in Qucs-S where asked, before you measure it.

Before you come

Pre-lab

Each team prints the report sheet, `LAB3-report.pdf`, and brings to the lab its shaded rows, like this, in Tables 1, 2 and 3, filled in by hand and in Qucs-S as Part A's Predict steps ask, with screenshots of the Qucs-S simulations on your laptop. Your grader checks them at the start of the session. Part B has no pre-lab.

Submitting

- **Part A** must be finished in the session: your grader signs off Checkpoints 1 and 2 before it ends, and accepts nothing of Part A later.
- **Part B**, M2, begins once Part A is signed off and is submitted after the lab: your M2 grader signs it off by Sunday 15 November, the first Sunday after the midterm week, at the latest. To submit, meet your M2 grader in their announced office hours or at another time you agree.
- **Your M2 grader:** Mikhail Nady Daoud for Group A, Dr. Asem Alaa for Group B and Sara Habo for Group C.
- **The report sheet** is due on Google Classroom by the same Sunday, 15 November.

Component values

The values and ratings given in this statement, of resistors, capacitors and every other part, are a starting point. If another value or part makes a circuit work better, a louder heartbeat or a steadier board 1′, you may use it: predict with the values given, then say in your report sheet, or for M2 in your radio guide, which part you changed, to what, and why.

The course page's guide shows how to install Qucs-S and ngspice: asem.alaa.page/ed/2026/eleg103/qucs-s. The 2N3904 is in the Libraries tab, under System Libraries, then Transistors; the 2N7000 under NMOSFETs.

Equipment and parts

Part A, in the lab

On the bench
Power supply: 12 V, then 15 V
Function generator
Oscilloscope and a probe
Digital multimeter
Breadboard and jumper wires
A stopwatch: your phone

Your own kit	Qty
2N3904 transistor	2
2N7000 MOSFET	1
Electret microphone capsule, two pins	1
A small plastic funnel, its mouth about 5 cm across, your own; tape	1
Wired headphone or earphone, 3.5 mm plug, your own; a 3.5 mm jack socket, or two crocodile clips	1
9.1 V Zener diode (1N4739A); red LED; push button	1 each
10 µF; 470 µF, 25 V	1; 3
10 kΩ	2
47 Ω, 330 Ω, 470 Ω, 1.2 kΩ, 2.7 kΩ, 3.9 kΩ, 22 kΩ, 39 kΩ, 82 kΩ, 100 kΩ	1 each

Part B, M2, after the lab

To build and measure it
Power supply at 4.5 V, or the cells
Function generator
Oscilloscope and a ×10 probe
Digital multimeter
LAB2's test loop

Your team's radio kit	Qty
M1's tank, from LAB2	1
Your team's breadboard, M1's, and jumper wires	1
2N3904 transistor	2
1N4148 diode; 1N5819 Schottky diode	1 each
1 kΩ potentiometer	1
Holder for three AA cells, and the cells	1
Insulated hook-up wire, for the link	1.5 m
10 µF; 1 µF; 100 nF; 10 nF; 3.3 nF	1; 1; 1; 2; 1
2 kΩ, 100 kΩ	2 each
220 Ω, 15 kΩ; for the test divider, 10 Ω and 10 kΩ	1 each

Part A1: An electronic stethoscope

A stethoscope carries the heart's sounds from the chest to the ear; this one does it electrically. In Figure 1, an electret microphone's capsule sits in the narrow end of a small funnel pressed to the chest. The capsule holds a field-effect transistor (FET) powered through 10 kΩ from 12 V, and feeds lecture L05's bench amplifier, Q_1, its emitter resistors bypassed by C_E. For the signal, the capsule is a source with that 10 kΩ as its own resistance. An earphone of 32 Ω straight on Q_1's collector would swamp its 3.9 kΩ, so a follower, Q_2, the common-collector stage of L05, stands between them.

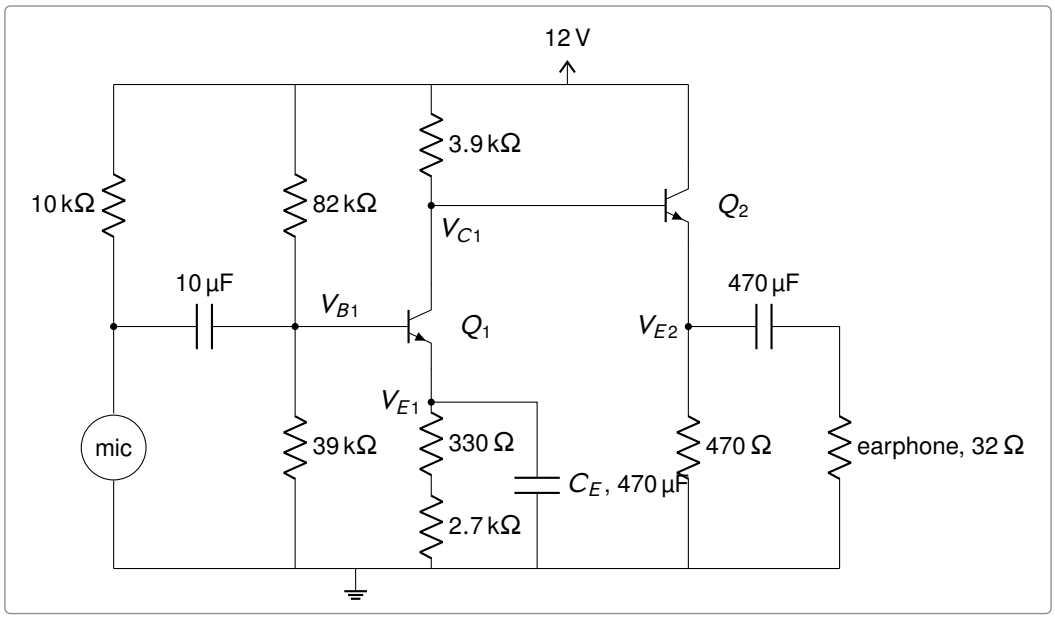


Figure 1: The stethoscope: the capsule in its funnel, lecture L05's bench amplifier (Q_1) and a follower (Q_2), both 2N3904s, into one earpiece of a 32 Ω headphone.

Predict (pre-lab)

Take $\beta = 100$, $V_{BE} = 0.7\,\text{V}$ and $V_T = 25\,\text{mV}$, as lecture L05 does, and leave out r_o. Take the earpiece as 32 Ω; if yours is marked otherwise, use its own value.

1. **DC.** With the capacitors open, replace the divider by its Thévenin source and find I_{E1}, V_{B1} and V_{E1}. Q_2's base current also flows in the 3.9 kΩ: write Kirchhoff's current law at Q_1's collector to find V_{C1}, then V_{E2} and Q_2's emitter current I_{E2}. Check that both transistors are active. Fill the Hand row of Table 1.
2. **The gain without the follower.** Find Q_1's g_m and r_π, the amplifier's input resistance R_{in}, the share of the capsule's signal that reaches the base through its 10 kΩ, and the gain from the capsule to the earphone when the earphone, through its 470 µF, hangs straight on Q_1's collector.
3. **The gain with the follower.** Find Q_2's input resistance and gain, as L05 found the common-collector stage's, and the gain from the capsule to the earphone. Fill the Hand row of Table 2.
4. **Simulate it in Qucs-S,** the capsule as an AC source behind 10 kΩ: a **dc simulation** for Table 1's Qucs-S row, and an **ac simulation** at 1 kHz for Table 2's, with and without the follower.

Build and measure

1. Build Figure 1 with the supply off, but leave the capsule and its 10 kΩ out for now. Each 470 µF has its stripe, its negative lead, towards ground on C_E and towards the earphone on the output. A 2N3904 seen from its flat face, leads down, has its emitter (E), base (B) and collector (C) from left to right. Clip the earphone's plug: the tip and the sleeve, the plug's base, give one earpiece of 32 Ω.
2. Set the supply to 12 V, its current limit near 50 mA, and switch it on. Measure V_{B1}, V_{E1}, V_{C1} and V_{E2}; I_{E2} is $V_{E2}/470\,\Omega$. Fill the Measured row of Table 1.
3. **The gain.** In place of the capsule, drive the 10 µF from the function generator through a 10 kΩ, which stands for the capsule's own; the 10 µF's stripe faces the generator, the lower side. Set a sine of 1 kHz and 40 mV peak to peak. Measure the generator's and the earphone's peak-to-peak voltages on the scope; their ratio is the gain. Then, with the supply off, move the output's 470 µF from Q_2's emitter to Q_1's collector, and measure the gain without the follower. Fill the Measured row of Table 2, and put the 470 µF back.

4. **Listen to a heart.** Switch off, put the capsule and its 10 kΩ in, its pin joined to the metal case towards ground, and turn the 10 µF round: the capsule's other pin sits above 7 V, so the stripe now faces Q_1's base. Tape the capsule into the funnel's narrow end, facing the wide end. Switch on first, then put the earpiece in, and take it out before you switch off: the output's 470 µF charges and empties through the earphone with a loud click. A light tap on the funnel should be clearly heard. Then press the funnel's rim firmly on the chest, just left of the breastbone, over a thin shirt: each beat comes as two short sounds.

Table 1: The stethoscope's operating point.

	V_{B1} (V)	V_{E1} (V)	V_{C1} (V)	V_{E2} (V)	I_{E2} (mA)
Hand					
Qucs-S					
Measured					

Table 2: The stethoscope's gain at 1 kHz, from the generator, behind 10 kΩ, to the earphone.

	Without the follower	With the follower
Hand		
Qucs-S		
Measured		

Checkpoint 1

Show your grader the stethoscope working, a heartbeat heard in the earpiece, and Tables 1 and 2.

Part A2: A defibrillator's ready light

A defibrillator charges a capacitor, then lights a lamp to say it is ready to shock. In Figure 2, 470 µF charges from 15 V through 22 kΩ. Until the capacitor passes 9.1 V, the Zener diode blocks and the 100 kΩ holds the 2N7000's gate at 0 V. Above it, the Zener conducts and the gate follows the capacitor, 9.1 V below it; once the gate passes the threshold V_{TH}, the MOSFET switches the ready LED on. The shock button empties the capacitor through 47 Ω.

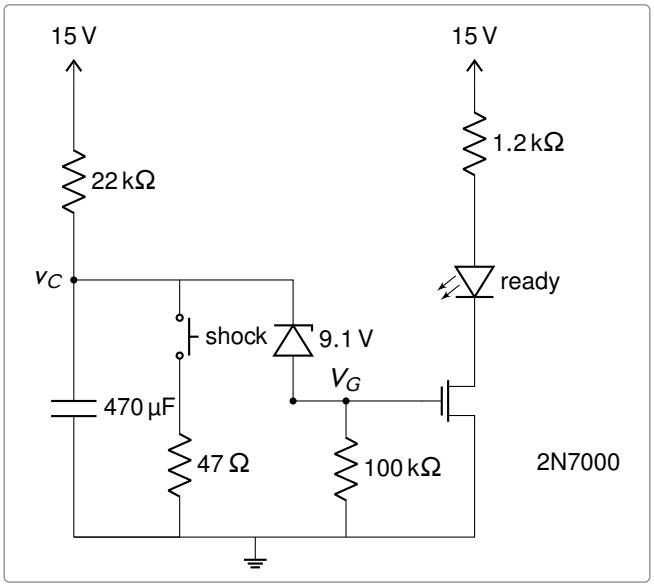


Figure 2: The ready light: v_C is the capacitor's voltage and V_G the gate's.

Predict (pre-lab)

Take $V_{TH} = 2.24\,\text{V}$ and $k_n = 211\,\text{mA V}^{-2}$, lecture L06's 2N7000, and the LED's 1.9 V.

1. **The first phase.** Below 9.1 V only the 22 kΩ feeds the capacitor: find the time constant τ and the time v_C takes to reach 9.1 V from 0 V.
2. **The second phase.** Above it, the Zener conducts and the 100 kΩ draws from the capacitor too. Find the Thévenin source the capacitor then sees, its new time constant τ_2, and the voltage at which it stops.
3. **The trip.** The LED lights once $V_G = v_C - 9.1\,\text{V}$ reaches V_{TH}: find that v_C and the time from switch-on.
4. **Lit.** When the capacitor has stopped, find V_{GS} and the LED's current: assume saturation, check it, and use the triode region if the check fails. Fill the Hand row of Table 3.
5. **Simulate it in Qucs-S**: a transient simulation from 0 s to 30 s, the capacitor starting empty. Read when the LED's current passes 1 mA, v_C then, and the LED's current at the end. Fill the Qucs-S row.

Build and measure

1. Build Figure 2 with the supply off. The capacitor's stripe goes to ground; the Zener's band, its cathode, to the capacitor; the LED's short lead to the drain. A 2N7000 seen from its flat face, leads down, has its source (S), gate (G) and drain (D) from left to right. Hold it by its body: static from a finger can destroy a MOSFET's gate.
2. Hold the shock button for a second to empty the capacitor. Put the multimeter across the capacitor, set the supply to 15 V, its current limit near 50 mA, and start your stopwatch as you switch it on. Note the time and v_C when the LED first glows.
3. A minute later, measure v_C, V_G, and the voltage across the 1.2 kΩ: the LED's current. Fill the Measured row of Table 3.
4. Press the shock button: the LED goes out at once. Release it, and time the second charge.
5. **Your own pair.** When the LED first glows, V_G is your MOSFET's threshold and $v_C - V_G$ your Zener's voltage. Measure both on a second charge: the Zener carries only tens of microamperes here, far below the current its 9.1 V is specified at.

Table 3: The ready light, from switch-on with the capacitor empty.

	τ, τ_2 (s)	v_C at the trip (V)	Time to light (s)	v_C at the end (V)	LED's current (mA)
Hand					
Qucs-S	—				
Measured	—				

Checkpoint 2

Show your grader the ready light coming on, its time against your prediction, the shock, and Table 3.

Part B: The radio's amplifier (M2)

M1, in LAB2, gave you a tank that picks one station and a detector that follows its peaks. You measured the detector's weakness there: its diode needs a few tenths of a volt before it conducts well, and a station arrives at the coil as millivolts or less. Board 1′ in Figure 3 amplifies the station first, at the station's own frequency, a radio frequency (RF): two CE stages, Q_1 and Q_2, each a 2N3904 with 2 kΩ from the supply to its collector, as

in lecture L05. A link of six turns, wound over M1's coil, takes the station off the tank; the 1 kΩ potentiometer across it sets how much reaches Q_1, the RF gain control that a strong station needs. Q_2's collector then drives its own detector, a 1N4148 with 15 kΩ and 3.3 nF, as M1's did.

Board 1′ has the radio's five connections: IN, where the link coil's live end connects; OUT, the sound, through a 1 µF capacitor; LEVEL, the detector's DC, taken before that capacitor, which a later milestone uses; V+, the supply, through a 1N5819 Schottky diode that blocks the current if the cells go in backwards; and ground (GND). On the breadboard each is a row of its own; the five-pin header that will carry them comes with the soldering, in M3. The board has three test points: test point 1 (TP1), the top of M1's tank; test point 2 (TP2), Q_2's collector, the amplified station; and test point 3 (TP3), the detector's output, which is also LEVEL.

Part B has no pre-lab: it begins once your grader has signed off Part A. Document M2 as the second section of your team's radio guide, the do-it-yourself (DIY) website that began with M1. Your M2 grader reads it for clarity, for ease of replication, for its illustrations, and for the challenges you met and how you worked round them.

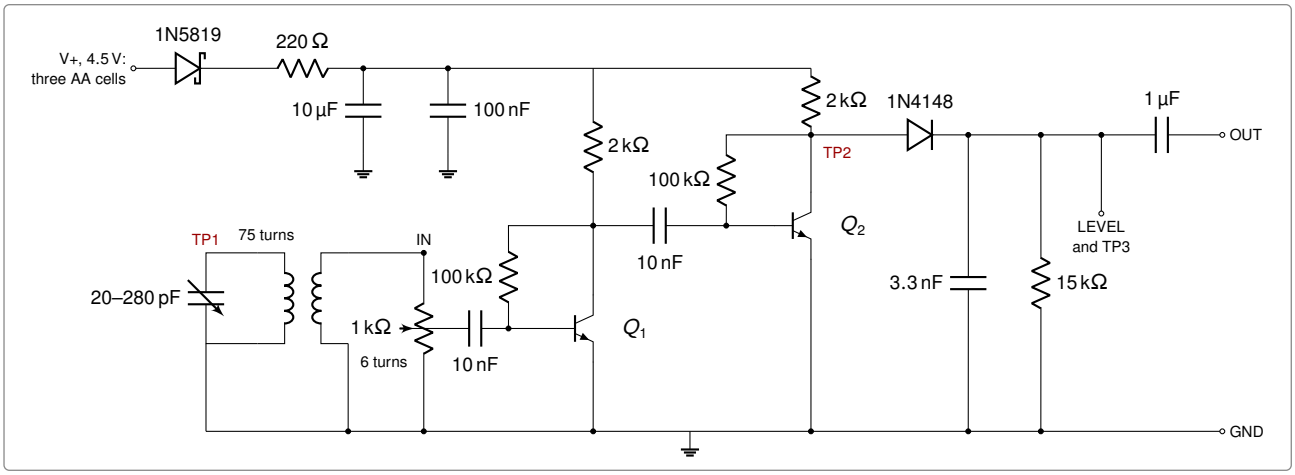


Figure 3: M2, board 1′, on M1's tank: two CE stages, each biased by 100 kΩ from collector to base, and a detector. TP2 is Q_2's collector.

How each stage is biased

Each stage sets its own bias with one resistor, $R_F = 100\,\text{k}\Omega$, from its collector to its base, in place of lecture L05's four resistors. Round the loop from the supply rail, V_R, through $R_C = 2\,\text{k}\Omega$ and R_F to the emitter, Kirchhoff's voltage law gives

$$V_R = R_C\,(I_C + I_B) + R_F\,I_B + V_{BE}, \qquad \text{so} \qquad I_E = \frac{V_R - V_{BE}}{R_C + R_F/(\beta+1)}, \qquad V_C = V_{BE} + R_F\,I_B.$$

The bias holds as L05's does: if I_C grows, V_C falls, and so does the base current through R_F. For the signal, R_F feeds part of the output back to the base; Qucs-S takes care of that.

Predict and build

Your team's radio kit for M2 is listed under Equipment and parts.

1. **Predict the bias by hand.** With $\beta = 100$, $V_{BE} = 0.7\,\text{V}$ and the rail at $V_R = 3.9\,\text{V}$ (the 1N5819 drops about 0.15 V at this small current, the 220 Ω about half a volt), find each stage's I_C and V_C. The stages are identical, so $V_{C1} = V_{C2}$ by hand.
2. **Predict the gain's ceiling.** With $g_m = I_C/V_T$ and $V_T = 25\,\text{mV}$, each stage alone could give $-g_m R_C$. Joined, each stage loads the one before it, and R_F feeds back part of the output, so the real gain is lower.
3. **Simulate it in Qucs-S.** Draw Figure 3 from IN onwards, with a 4.35 V supply standing for the cells and the 1N5819, which Qucs-S's library lacks, and drive IN from an AC source through a 10 kΩ into 10 Ω to ground, the test divider; put 15 pF from TP2 to ground for the scope's probe. Run a **dc simulation** for the

operating point and an **ac simulation** from 300 kHz to 3 MHz, and read the gain from IN to TP2 at 531 kHz, 1 MHz and 1.6 MHz, the ends and the middle of the medium-wave band. Fill the Hand and Qucs-S rows of Tables 4 and 5.

4. **Lay out the board** on your breadboard in the order of Figure 3, in a line: IN and the potentiometer at one end, then Q_1, then Q_2, and the detector and OUT at the other end, with at least ten rows between Q_1's base and Q_2's collector. Two neighbouring rows of a breadboard couple through a few picofarads, and less than one picofarad from Q_2's collector back to Q_1's base can make the board oscillate. Keep every lead short and flat on the board, and run no wire from the output end back past Q_1. Run GND along one of the breadboard's rails and the supply, after the 1N5819 and the 220 Ω, along the other; some breadboards break their rails halfway along, so bridge the break. Put the 10 µF and the 100 nF where the supply enters its rail, the 10 µF's stripe, its negative lead, towards GND. A 2N3904 seen from its flat face, leads down, has its emitter (E), base (B) and collector (C) from left to right.

5. **Build it,** then, before any power, check every part's rows against Figure 3, and check with the multimeter's continuity test that V+ does not reach GND. Mark V+, and check the cells' polarity before you put them in: the 1N5819 blocks a reversed supply, so a board that draws no current may simply have its cells in backwards.

6. **Wind the link:** six turns of the hook-up wire, side by side, over the end of M1's coil that goes to ground, in the same direction as the coil. Twist the link's two ends together and connect them to IN and GND. M1's detector is no longer needed on the tank: board 1′ has its own.

Measure

Power the board from its three AA cells or from a supply set to 4.5 V with its current limit near 20 mA. For steps 2 to 5, take the link off IN and drive IN from the generator through the test divider, 10 kΩ from the generator to IN and 10 Ω from IN to GND: the divider passes about 1 mV for each volt from the generator, and the generator never needs more than 1 V. From step 3 on, turn the potentiometer fully up. Read TP2 with the probe and the channel both on ×10.

1. **The operating point.** With no signal, measure the current the board draws, the rail after the 220 Ω, V_{C1}, V_{C2} (TP2's DC) and TP3's DC, the detector at rest. Fill the Measured row of Table 4.

2. **No oscillation.** With the generator connected but switched off, turn the potentiometer slowly from one end to the other: at every setting, TP2 on the scope should sit at its DC level with only a few millivolts of noise. A steady sine of a volt or more, at a few megahertz or above, means the board oscillates: move Q_2's collector and the detector farther from Q_1's base and IN, and shorten the leads round them.

3. **The gain.** At 1 MHz, set the generator to 0.1 V and read TP2's amplitude; then at 531 kHz and 1.6 MHz. The gain is TP2's amplitude over IN's, the generator's amplitude divided by the divider's ratio. Fill the Measured row of Table 5.

4. **The detector.** At 1 MHz, record TP2's amplitude and how far TP3 rises above its rest, at 0.05, 0.1, 0.2, 0.3, 0.5 and 1 V from the generator. Fill Table 6 and compare it with M1's: board 1′'s input here is about a thousandth of what M1's detector needed.

5. **The RF gain control.** At 1 V from the generator, TP2's peaks flatten: turn the potentiometer down until they are round again, and note the setting.

6. **On M1's tank.** Join the link to IN again, the board at least 15 cm from the coil and the tuning capacitor, its TP2 and detector end farthest from them. With the generator off, TP2 should again show only noise, at every setting of the potentiometer: a sine of a volt or so that the tuning capacitor moves means the whole radio oscillates, through a fraction of a picofarad between the board and the tank's top. Move the board away, or turn the potentiometer down. Then drive LAB2's test loop from the generator at 0.1 V, about 10 cm from your coil and on its axis, and tune the capacitor for the largest rise of TP3, turning the potentiometer down if TP2's peaks flatten. Then, with the potentiometer fully up, move the loop away until TP3 rises by

only about 10 mV, and measure that distance; in LAB2, M1 alone needed the loop a few centimetres away. Fill Table 7.

7. **A station, if you can (not required for the sign-off).** By a window, with the coil upright, connect OUT to a powered speaker or your laptop's sound card, tune slowly across the band, and list what you hear.

Table 4: M2's operating point. Hand: $\beta = 100$, $V_{BE} = 0.7$ V, the rail at 3.9 V.

	Supply current (mA)	Rail (V)	V_{C1} (V)	V_{C2}, TP2 (V)	TP3 at rest (V)
Hand		3.9			
Qucs-S					
Measured					

Table 5: M2's gain from IN to TP2, the generator at 0.1 V.

	531 kHz	1 MHz	1.6 MHz
Qucs-S			
Measured			

Table 6: The detector at 1 MHz: TP2's amplitude, and TP3's rise above its rest, at six settings of the generator.

Generator (V)	IN (µV)	TP2's amplitude (mV)	TP3's rise (mV)
0.05			
0.1			
0.2			
0.3			
0.5			
1			

Table 7: M1 and M2 on the test loop, the generator at 0.1 V.

Tuned to (kHz)	Loop's distance for a 10 mV rise (cm)	Potentiometer's setting

Checkpoint 3: M2 signed off

By Sunday 15 November at the latest, show your M2 grader the board on its breadboard, the clean TP2 of step 2, Tables 4 to 7, and the board on M1's tank answering the test loop. Keep M1 and M2: the next milestones build on them.

Supplementary reading: M2 in context. No step of the lab depends on it.

Where M2 sits in a radio. A station reaches your coil as millivolts, often as microvolts, and a diode cannot follow so small a signal. So radios amplify the station first, at its own frequency, and detect it afterwards (Figure 4): a tuned circuit, gain, then a detector, the *tuned radio-frequency* receiver of the 1920s. Board 1′ adds a second idea: its diode rests just conducting, held there by Q_2's collector, so that even a small signal moves it along the bend of its curve, lecture L02's diode with a small signal on its bias. Gain at radio frequencies has one danger: any path from the output back to the input, a few tenths of a picofarad between two wires, turns the amplifier into an oscillator. That is why M2 is laid out in a line, its output far from its input.

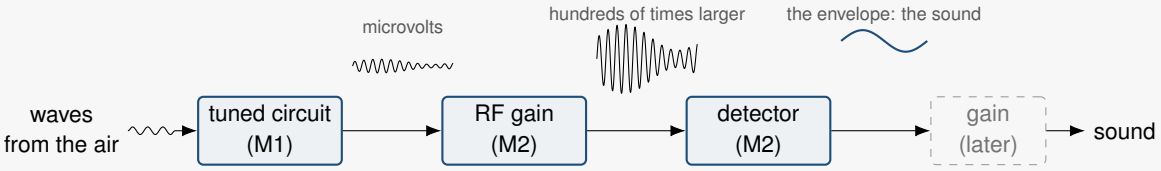


Figure 4: M2 in a radio: the gain comes before the detector, at the station's own frequency.

Who made it. In 1906 Lee de Forest put a third electrode, a grid, inside a vacuum-tube detector: the Audion, the first device that could amplify. In 1912 Edwin Howard Armstrong, a student at Columbia University, found that feeding part of the Audion's output back to its input multiplied its gain many times over; he patented this *regenerative* receiver in 1914 (Figure 5). Pushed further, the same feedback made the Audion oscillate, so his circuit could also generate radio waves: the fault that M2's layout guards against. The radios of the 1920s stacked several tuned stages of gain before the detector, until the *superheterodyne*, which Armstrong built in 1918, took their place. After the transistor's invention in 1947, the Regency TR-1 of 1954 put a whole radio, with four transistors, in a pocket.

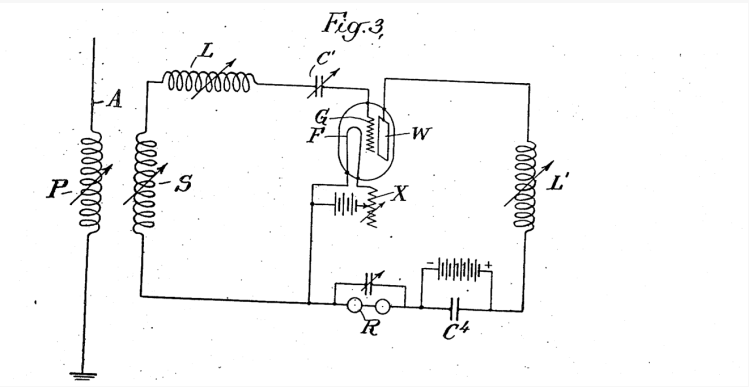
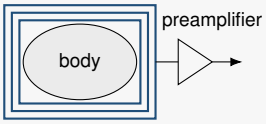
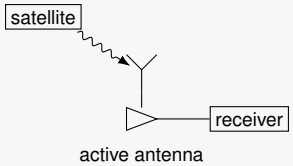


Figure 5: Armstrong's *Wireless receiving system*, United States patent 1,113,149 (1914), Fig. 3: the antenna (A) feeds a tuned circuit (S, L, C′) on the Audion's grid (G); the coil L′ in its plate circuit (W) returns part of the output to the input, and the telephone (R) gives the sound.

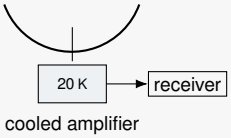
Beyond radio. Wherever a faint signal must cross a cable or a detector, an amplifier sits right at the sensor, as M2 sits right at M1's tank (Figure 6).



(a) Medicine: magnetic resonance imaging (MRI). The protons in the body ring at 42.58 MHz per tesla, 64 MHz in a 1.5 T scanner, with a signal of microvolts. Each receive coil, tuned like M1's tank, has its own low-noise amplifier at the coil.



(b) Navigation: the Global Positioning System (GPS). A satellite's signal reaches the ground weaker than the receiver's own noise. An amplifier inside the antenna raises it before the cable to the receiver can add noise of its own.



(c) Science: radio telescopes. The signals from space are so faint that the first amplifier, at the dish's focus, is cooled to about 20 K, −253 °C, to keep its own noise below them.

Figure 6: M2's idea, an amplifier at the sensor, beyond radio.

Report

Each team fills in the report sheet, `LAB3-report.pdf`, or its LaTeX in `LAB3-report-latex.zip`, and submits it as the sheet says. M2 goes into your radio guide instead.